$$\begin{tikzcd}
F(A) \arrow[r, "F(f)"] \arrow[d, "\eta_A"'] & F(B) \arrow[d, "\eta_B"] \\
G(A) \arrow[r, "G(f)"'] & G(B)
\end{tikzcd}$$

Figure 1: This would be better if you could put a box around it and resize it

$$\boxed{\begin{tikzcd}
F(A) \arrow[r, "F(f)"] \arrow[d, "\eta_A"'] & F(B) \arrow[d, "\eta_B"] \\
G(A) \arrow[r, "G(f)"'] & G(B)
\end{tikzcd}}$$

Figure 2: Natural Transformation: $\eta : F \to G$

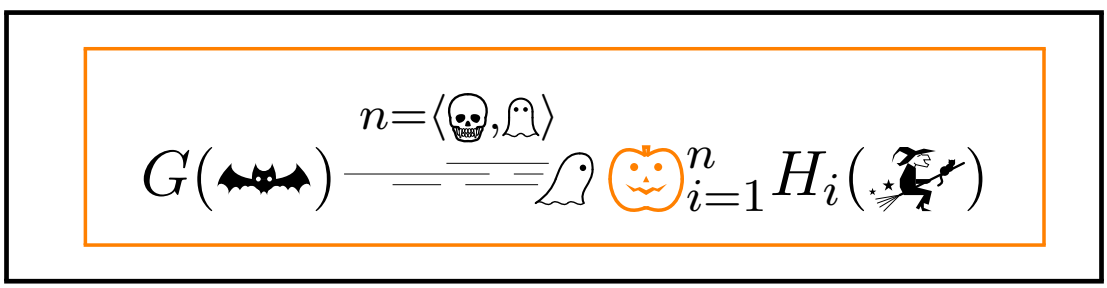

Figure 3: Happy Halloween!

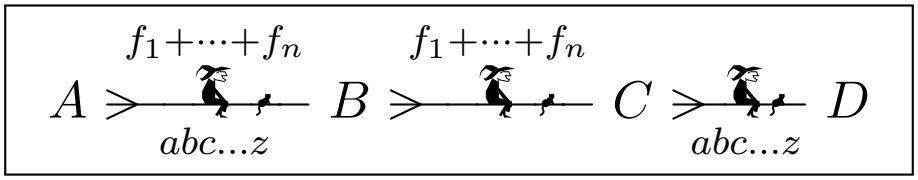

Figure 4: Happy Halloween!

0.6249999121401877

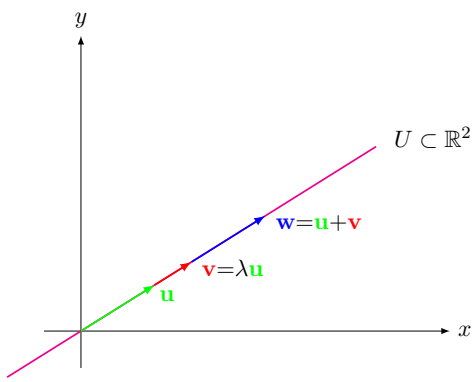

Figure 5: The line U is a linear subspace of $\mathbb{R}^2$

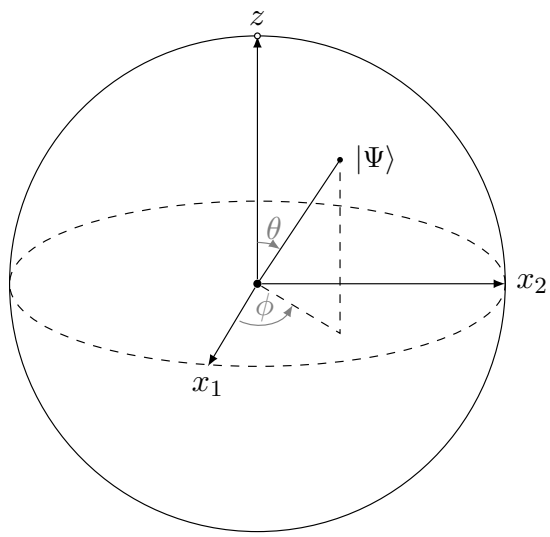

Figure 6: The Bloch Sphere

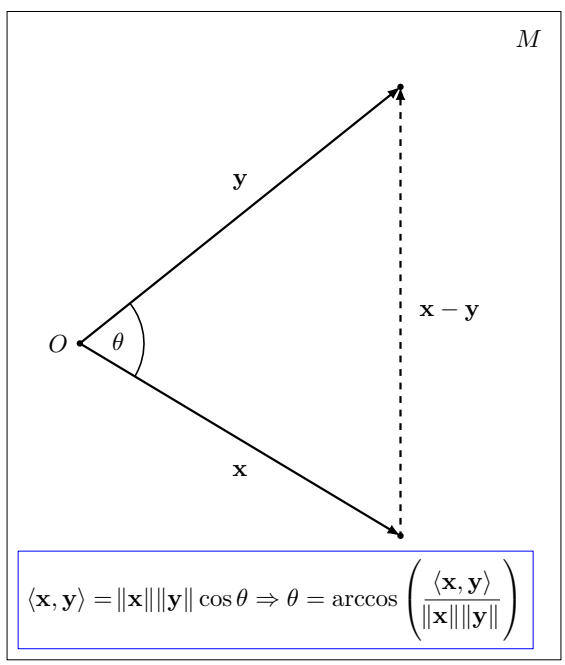

Figure 7: Geometric interpretation of $\langle \mathbf{x}, \mathbf{y} \rangle$

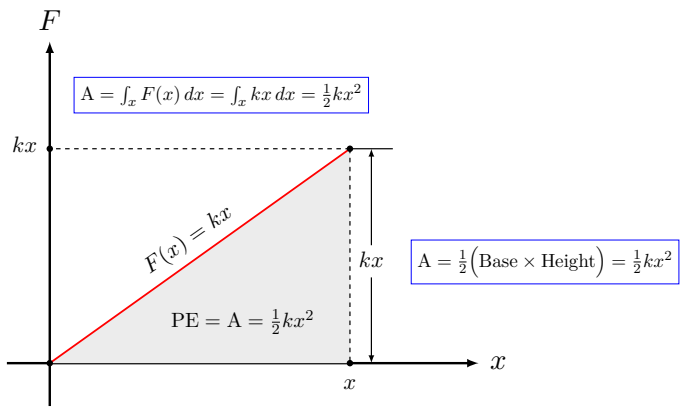

Figure 8: Potential Energy of a Simple Harmonic Oscillator

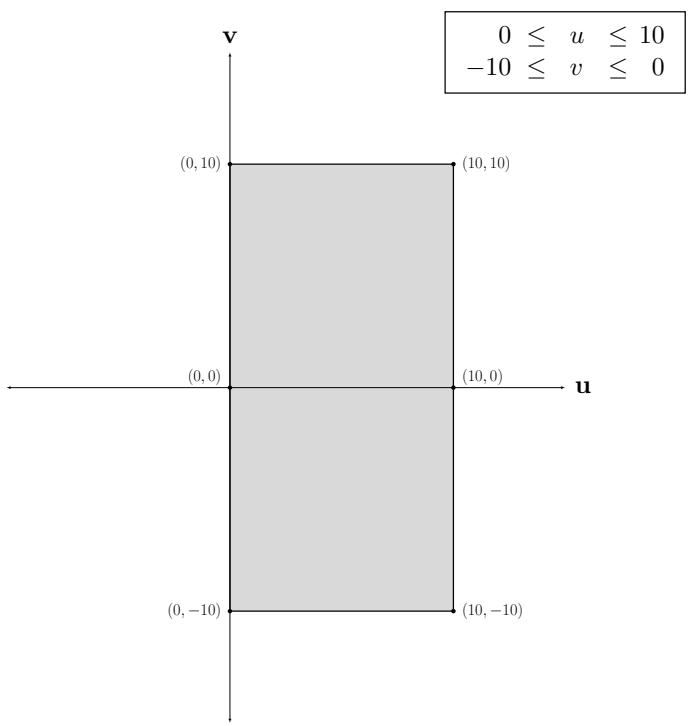

Figure 9: The transformed parallelogram is a rectangle in the uv-plane

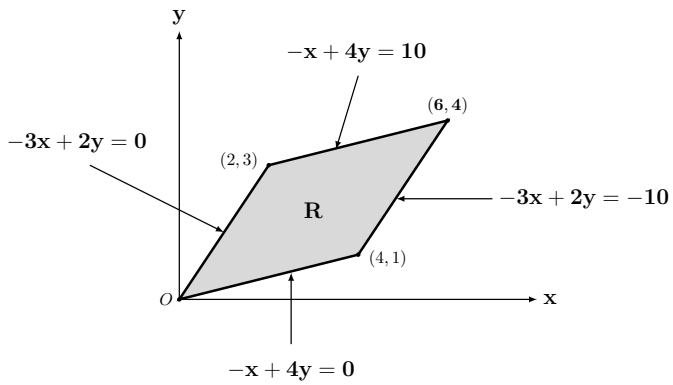

Figure 10: Equations of the Sides of the Parallelogram

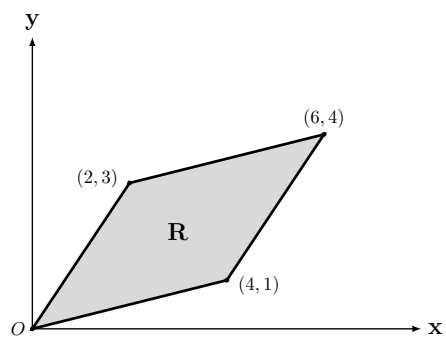

Figure 11: Parallelogram in the xy-plane

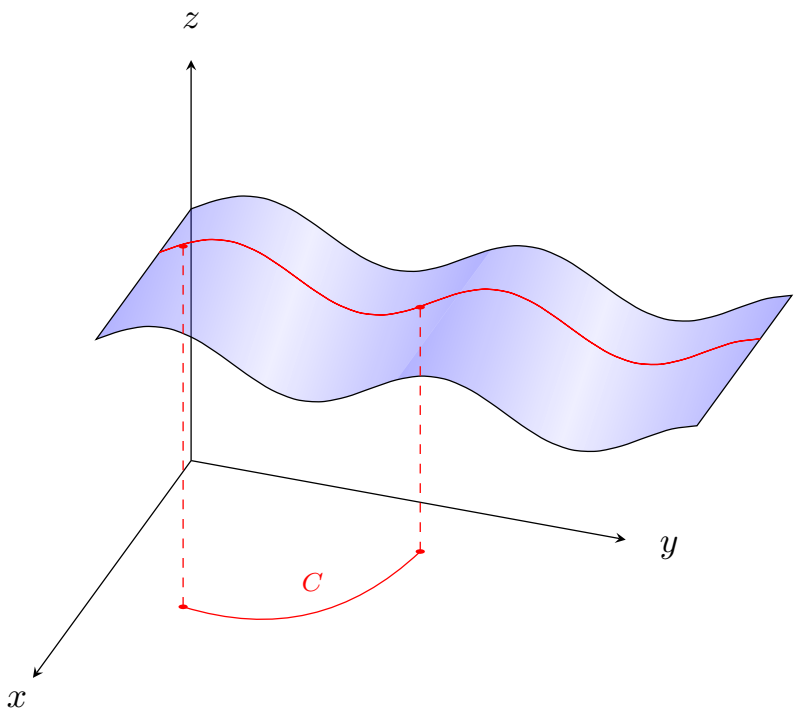

Figure 12: 3D

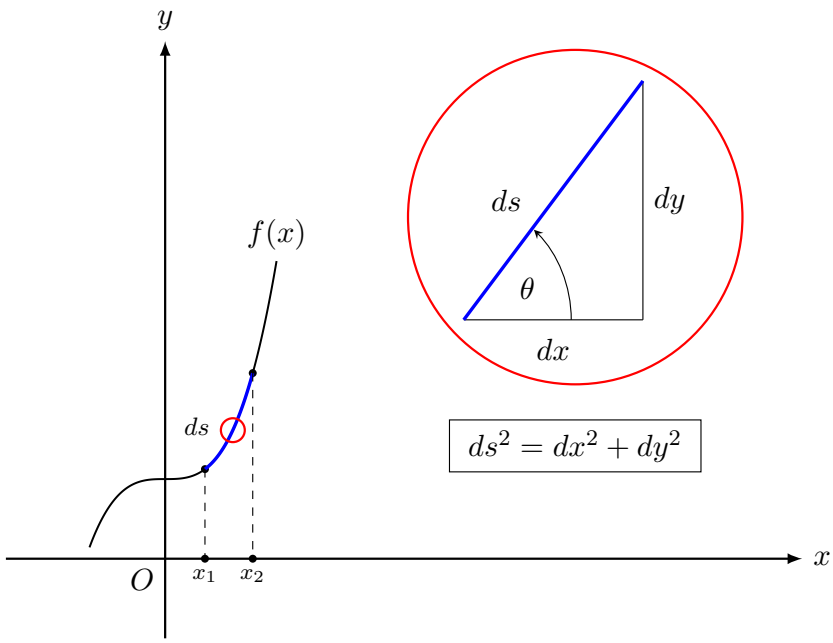

Figure 13: $f(x)$, ds and the Pythagorean Theorem

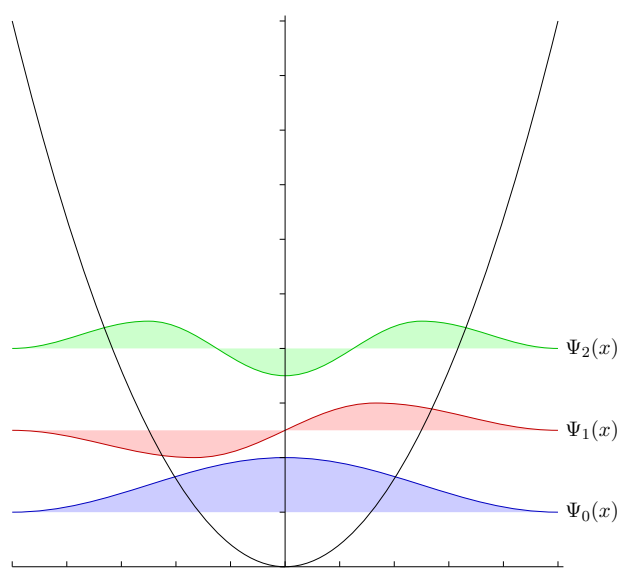

Figure 14: Wave Functions

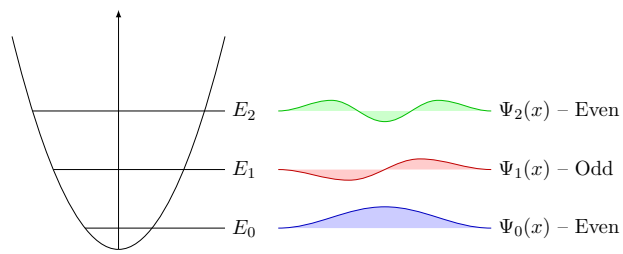

Figure 15: Wave Functions

9^2	$=$	81	$\longrightarrow$	$8+1$	$=$	9
45^2	$=$	2025	$\longrightarrow$	$20+25$	$=$	45
703^2	$=$	494209	$\longrightarrow$	$494+209$	$=$	703
7777^2	$=$	60481729	$\longrightarrow$	$6048+1729$	$=$	7777
857143^2	$=$	734694122449	$\longrightarrow$	$734694+122449$	$=$	857143

Figure 16: A Few Example Kaprekar Numbers

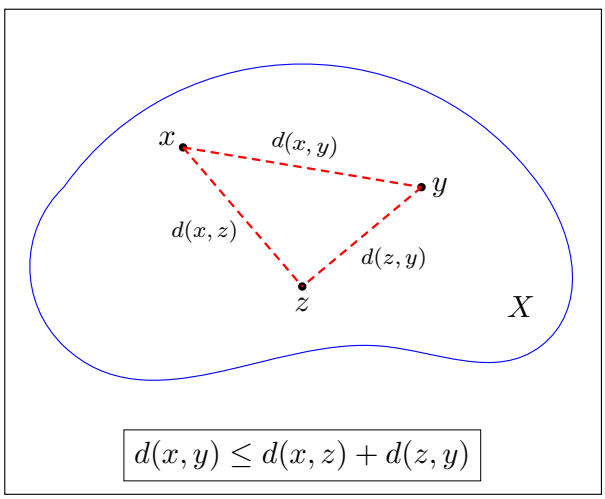

Figure 17: The Triangle Inequality

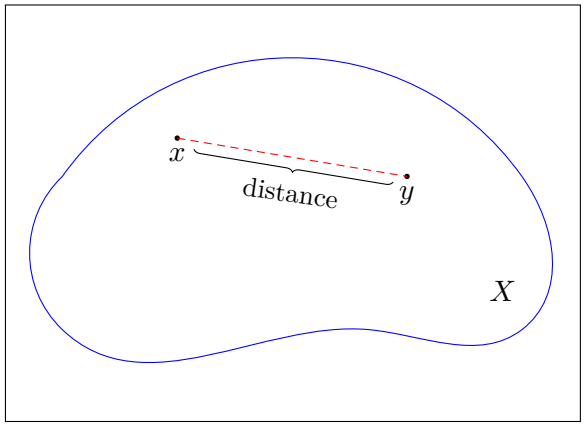

Figure 18: X, x and y

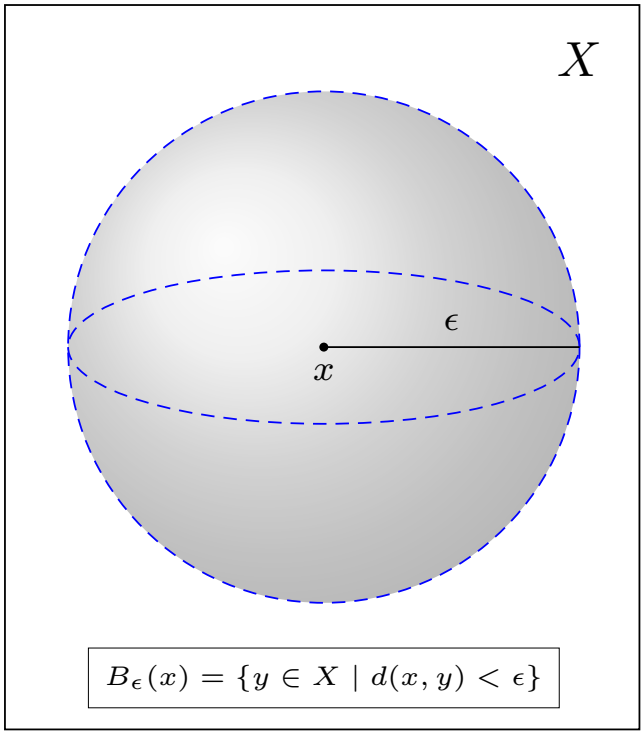

Figure 19: $B_\epsilon(x)$ is an open epsilon ball centered at x with radius ϵ

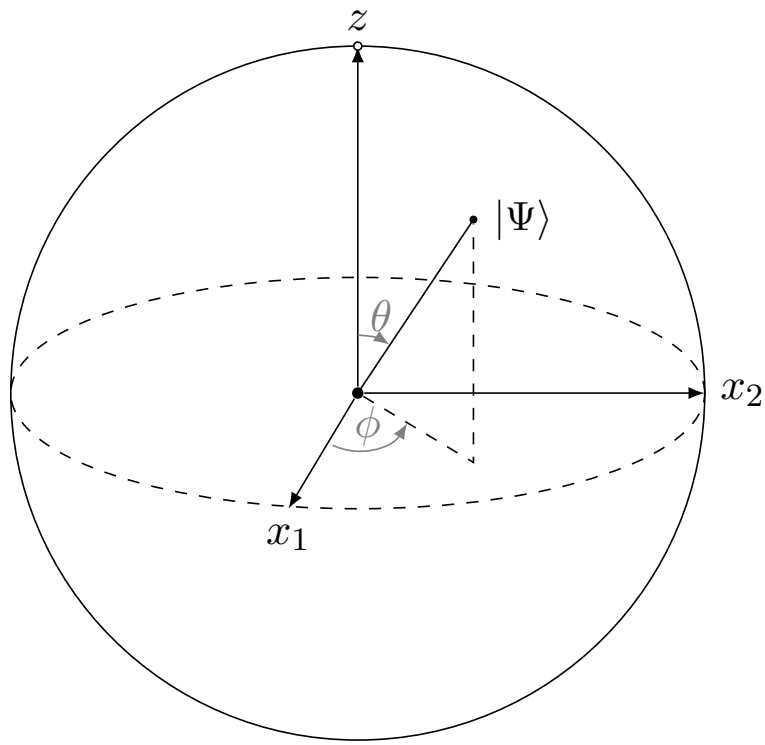

Figure 20: The Bloch Sphere

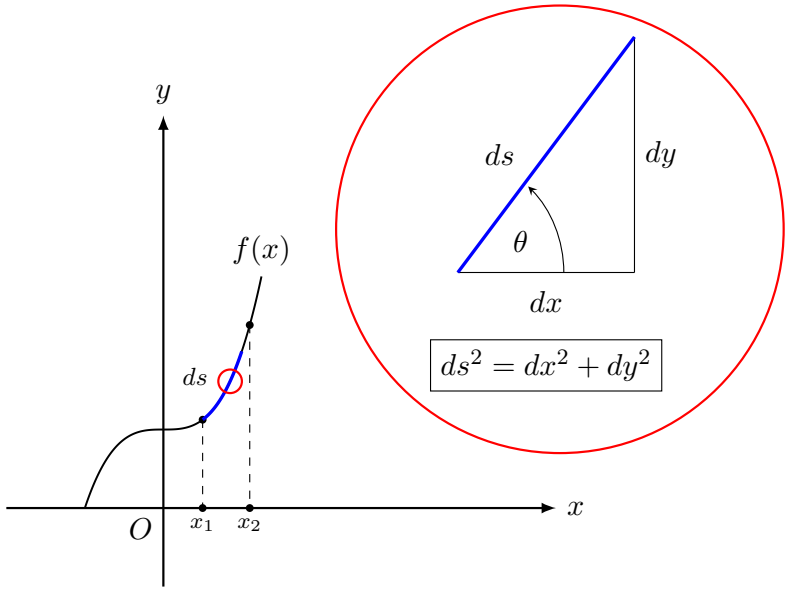

Figure 21: $f(x)$, ds and the Pythagorean Theorem

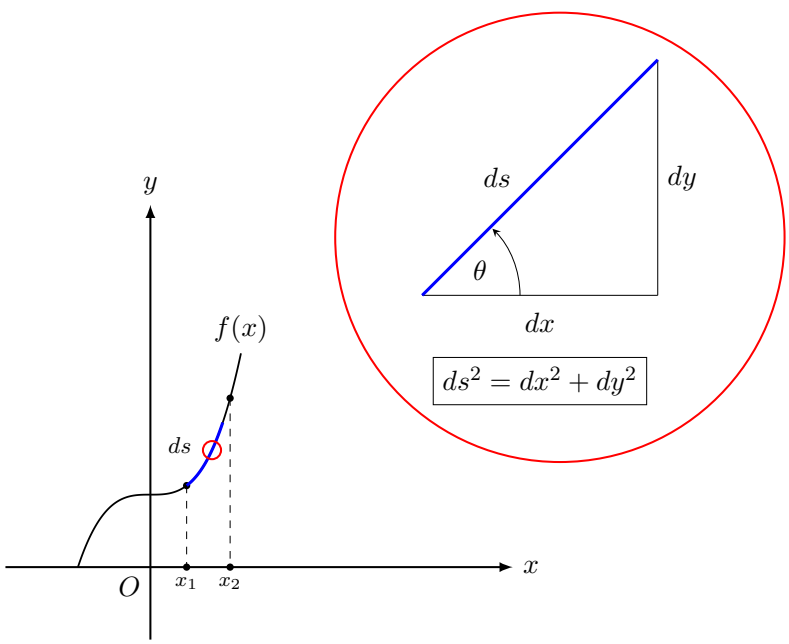

Figure 22: $f(x)$, ds and the Pythagorean Theorem